

Supplementary Information for legoESM: a composable, differentiable, multi-scale Earth system model built with AI agents

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This Supplement provides (§1) a self-contained description of the governing equations and numerical methods; (§2) dynamical-core verification; (§3) composability details; (§4) the large-eddy and cloud-resolving suites; (§5) ocean validation; (§6) performance and scaling; (§7) accessibility and a code example; and (§8) operator-level physical-consistency tests. A complete treatment is in the legoESM Scientific Guide that ships with the source code¹.

S1 Model description: governing equations and numerics

legoESM is a federation of packages sharing one namespace under a one-way dependency graph (`core` $\prec$ `{atmosphere, ocean, land, ice}` $\prec$ `{coupler, ml, tools}`; main-text Fig. 1). All code is pure-functional JAX; time integration uses `lax.scan`; spectral cores use `float64/complex128` while finite-volume cores may use `float32`.

S1.1 Grids and vertical coordinates

Cubed sphere. Six faces are mapped to the sphere by an equiangular gnomonic projection^{2,3}. With equiangular coordinates $(\alpha_x, \alpha_y) \in [-\pi/4, \pi/4]^2$ the cell area on a Cn face is

$$\Delta A = \frac{R^2 \Delta \alpha^2}{\cos^2 \alpha_x \cos^2 \alpha_y D^3}, \quad D = \sqrt{1 + \tan^2 \alpha_x + \tan^2 \alpha_y}, \quad \Delta \alpha = \frac{\pi}{2n}, \quad (\text{S1})$$

and integrating over the six faces recovers $4\pi R^2$ to round-off (discrete area conservation). Vector halo exchange rotates to geographic components across panel seams; a duo-grid treatment of the edges removes the residual differential-operator imprint⁴.

Spectral/Gaussian. Fields are represented by spherical-harmonic coefficients with triangular truncation at total wavenumber $n_{\max}$ ⁵. Analysis is an FFT in longitude followed by Gaussian quadrature in latitude with the fully normalized associated Legendre functions P_n^m . The Laplacian is diagonal, $\nabla^2 \hat{f}_{nm} = -n(n+1)/a^2 \hat{f}_{nm}$, giving exact, grid-imprint-free linear operators and hyperdiffusion $-\nu[n(n+1)/a^2]^p$.

Voronoi/MPAS. An unstructured centroidal Voronoi tessellation with mimetic TRiSK operators^{6–8}: scalars at cell centres, normal velocity at edges, vorticity at vertices, with

$$(\nabla \cdot \mathbf{u})_c = \frac{1}{A_c} \sum_e n_{e,c} u_e l_e, \quad (\nabla \phi)_e = \frac{\phi_{c_2} - \phi_{c_1}}{d_e}, \quad (\nabla \times \mathbf{u})_v = \frac{1}{A_v} \sum_e t_{e,v} u_e d_e. \quad (\text{S2})$$

Vertical coordinates. The atmosphere uses sigma $\sigma = p/p_s$ or hybrid sigma–pressure $p(k) = A(k)p_{\text{ref}} + B(k)p_s$; geopotential uses the energy-conserving Simmons–Burridge discretization⁹

$$\Phi_k = \Phi_s + R_d \sum_{k'=k+1}^{N-1} T_{k'} \ln \frac{\sigma_{k'+1/2}}{\sigma_{k'-1/2}} + \alpha_k R_d T_k, \quad \alpha_k = 1 - \frac{\sigma_{k-1/2}}{\Delta \sigma_k} \ln \frac{\sigma_{k+1/2}}{\sigma_{k-1/2}}. \quad (\text{S3})$$

The ocean uses a dynamic z^* coordinate with free surface η and layer Jacobian $J(\eta) = (\eta + H)/H_{\max}$.

S1.2 Atmospheric dynamics

Shallow water (vector-invariant).

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h \mathbf{v}), \quad \frac{\partial u}{\partial t} = (\zeta + f)v - \frac{\partial B}{\partial x} + D_u, \quad \frac{\partial v}{\partial t} = -(\zeta + f)u - \frac{\partial B}{\partial y} + D_v, \quad (\text{S4})$$

with Bernoulli function $B = \frac{1}{2}(u^2 + v^2) + g(h + h_s)$ and relative vorticity ζ . On the cubed sphere the FV3-style C-D grid carries momentum on the D-grid and diagnoses C-grid mass fluxes, with PPM transport^{3,10,11}; the circulation-based corner vorticity removes the Hollingsworth–Källberg instability.

Hydrostatic primitive equations (σ).

$$\frac{\partial p_s}{\partial t} = -\int_0^1 \nabla \cdot (p_s \mathbf{v}) d\sigma, \quad (\text{S5})$$

$$\frac{\partial u}{\partial t} = (\zeta + f)v - \frac{\partial B}{\partial x} - R_d T \frac{\partial \ln p_s}{\partial x} + D_u, \quad (\text{S6})$$

$$\frac{\partial T}{\partial t} = -\nabla \cdot (T \mathbf{v}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \kappa T \frac{\omega}{p} + Q, \quad (\text{S7})$$

with $\kappa = R_d/c_p$ and diabatic heating Q from physics. The spectral core advances the vorticity–divergence form with the fast gravity-wave terms treated semi-implicitly¹², yielding a per-wavenumber Helmholtz solve pre-factored at initialization. The nonhydrostatic core integrates the fully compressible Euler equations on a terrain-following height coordinate for CRM/LES use.

Coupling. legoESM uses time-split coupling: the dynamical core updates the state, then the physics pipeline applies parameterizations in the order radiation $\rightarrow$ deep convection $\rightarrow$ shallow convection/large-scale condensation $\rightarrow$ microphysics $\rightarrow$ boundary-layer turbulence $\rightarrow$ gravity-wave drag $\rightarrow$ surface fluxes, followed by conservation fixers and diagnostics. A grid-agnostic column adapter reshapes any grid’s 3D state to $(n_{\text{col}}, n_{\text{lev}})$ for physics.

S1.3 Atmospheric physics (selected closures)

Radiation is either gray (Frierson-style¹³) or correlated- k RRTMGP¹⁴. Mass-flux convection (Tiedtke¹⁵, Zhang–McFarlane¹⁶, Bechtold, Kain–Fritsch, Emanuel, Kuo, EDMF, SBM) shares a unified updraft core. Surface exchange uses Monin–Obukhov similarity; boundary-layer mixing offers Louis, Holtlag–Boville, YSU, EDMF, TKE and Smagorinsky¹⁷ options. Warm-rain microphysics ranges from Kessler to two-moment schemes and a super-droplet method¹⁸. Each scheme is a factory returning `physics_fn(state, grid, coord)  $\rightarrow$  tendencies`, and the combined tendency is the exact sum of the enabled processes.

S1.4 Ocean

The ocean solves the hydrostatic Boussinesq primitive equations with a split-explicit free-surface barotropic solver¹⁹, KPP vertical mixing²⁰, GM–Redi isopycnal/eddy mixing²¹, and a TEOS-10-style polynomial equation of state. Tracers obey $\partial_t(h\theta) + \nabla \cdot (h\mathbf{u}\theta) = \nabla \cdot (h\mathbf{K}\nabla\theta) + \mathcal{S}_\theta$, with h the layer thickness and $\mathbf{K}$ the (rotated) diffusivity tensor. The barotropic subsystem is advanced either implicitly (Crank–Nicolson) or by a SOTA split-explicit substep with local CFL clamping; both conserve volume and tracers to the gated tolerance.

S1.5 Land, sea ice, coupler, ML

The multi-layer land model solves the mixed-form Richards equation²² for soil moisture, Johansen thermal diffusion for soil temperature, Farquhar–Ball–Berry photosynthesis²³ with stomatal conductance, and a DALEC-style carbon cycle. Sea ice uses EVP/mEVP rheology²⁴ with a multi-category thickness distribution and energy-conserving thermodynamics. The coupler exchanges area-weighted surface fluxes over sub-grid tiles with a Monin–Obukhov surface layer. The `ml` package provides spherical Fourier neural operators²⁵ (built on Fourier neural operators²⁶), conservation-aware losses, differentiable training drivers, and variational (4D-Var) data assimilation that differentiates through the compiled time-stepping kernel.

S2 Dynamical-core verification

legoESM is gated on the standard sphere test set^{27–29}. Figure S2 shows cosine-bell solid-body advection (Williamson case 1) on the cubed sphere; the bell traverses panel edges without distortion. Figure S3 shows the Williamson 5 (flow over an isolated mountain) and Williamson 6 (Rossby–Haurwitz wave) tests on the cubed sphere. The Jablonowski–Williamson baroclinic wave is shown for surface pressure and near-surface wind speed in Fig. S4. Crucially, because the equation set and

the numerical discretization are independent choices in legoESM, the same shallow-water problem can be solved on every grid: Fig. S1 shows that the cubed-sphere, lat–lon, icosahedral and spectral cores give visually indistinguishable solutions for the Rossby–Haurwitz wave and for flow over a mountain. Mass is conserved to round-off; energy and potential enstrophy budgets close to their scheme-dependent tolerances.

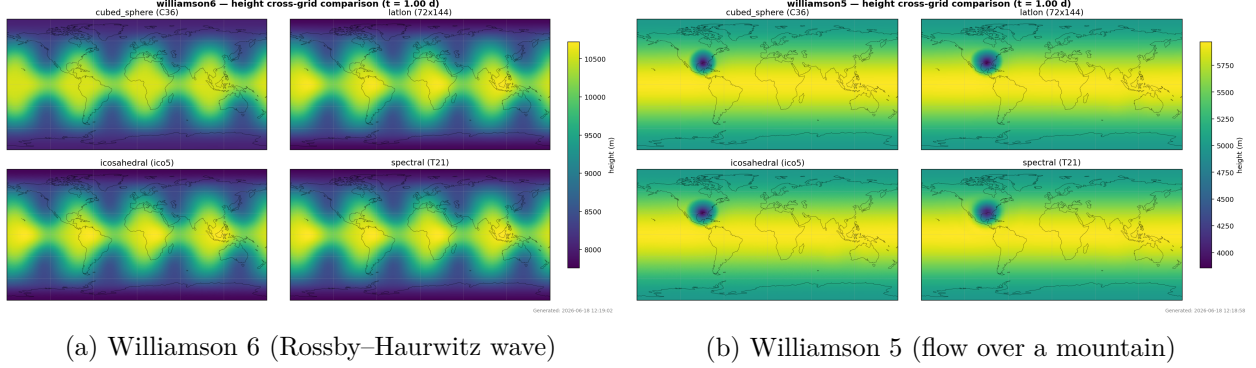


Figure S1: **The same shallow-water equations solved on four grids.** Height field for (a) the Rossby–Haurwitz wave (Williamson 6) and (b) flow over an isolated mountain (Williamson 5), each integrated on the cubed-sphere (C36), lat–lon (72×144), icosahedral (level 5) and spectral (T21) discretizations. The solutions are visually indistinguishable across grids, showing that the physical equation set and the numerical method are orthogonal choices²⁷.

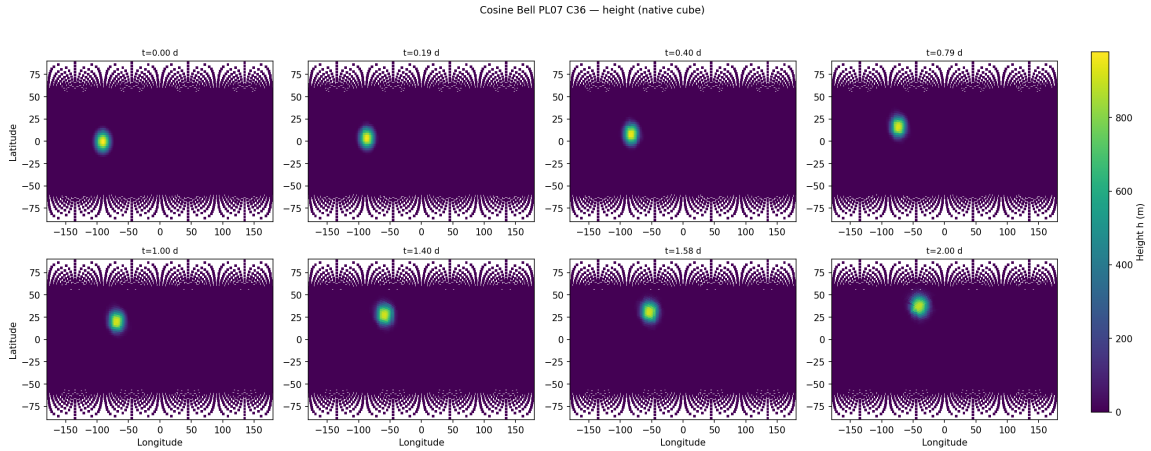


Figure S2: Cosine-bell advection (Williamson case 1) on the cubed sphere (C36), shown in the lat–lon projection at successive times. The tracer maintains its shape and amplitude as it crosses cube-panel edges.

S3 Composability

Composability operates on four orthogonal axes (main-text Fig. 1): grid, complexity, physics and fidelity. Physics composition is additive to machine precision, $\text{tendency}(A + B + C) = \text{tendency}(A) + \text{tendency}(B) + \text{tendency}(C)$, which is asserted by tests; a process is disabled by setting its scheme to "none", and every factory raises on an unknown scheme rather than silently

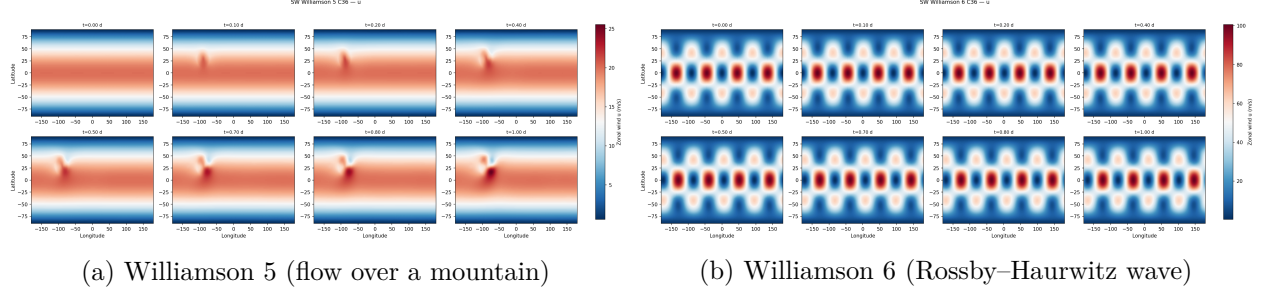


Figure S3: Shallow-water Williamson 5 and 6 tests on the cubed sphere (C36).

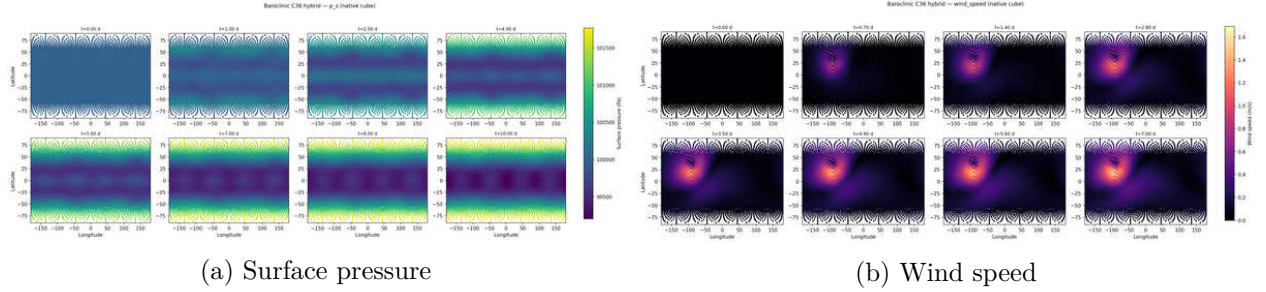


Figure S4: Jablonowski-Williamson baroclinic-wave test, hydrostatic cubed-sphere C36, days 0-10. The instability grows at the expected rate and location.

defaulting. An entire component can be swapped the same way: in a short atmospheric integration the same slot accepts (i) the classical physics suite, (ii) a column neural network, or (iii) a full SFNO dynamical core; and the ocean can be run as prescribed SST, a slab mixed layer, or a full three-dimensional model from one driver. Table S1 lists the dynamical cores; Tables S2-S3 list the physics and ocean/land/ice schemes.

Table S1: Dynamical cores by equation set and discretization family. The discretizations follow FV3-style cubed-sphere finite volume with PPM transport^{3,11} and a duo-grid panel-edge treatment⁴, lat-lon C-grid finite volume³⁰, semi-implicit spectral transforms^{5,12}, and mimetic TRiSK operators on Voronoi/MPAS meshes⁶⁻⁸.

Discretization	Shallow water	Hydrostatic PE	Nonhydrostatic
Cubed-sphere C-D (FV3)	✓	✓	✓
Lat-lon C-grid (FV)	✓	✓	—
Gaussian/spectral	✓	✓	✓
Voronoi/MPAS (TRiSK)	✓	✓	✓
SFNO (learned)	✓	✓	—
Plane (LES/CRM)	—	—	✓

S4 Multi-scale: large-eddy and cloud-resolving suites

On a doubly periodic plane with the nonhydrostatic core and a scale-dependent dynamic Smagorinsky closure, legoESM reproduces canonical LES cases: GABLS1³¹ (main-text Fig. 4a), the Ekman layer, the Wangara diurnal cycle, and the DYCOMS-II³² and RICO³³ cloud-topped boundary layers

Table S2: Atmospheric physics processes and representative schemes (each composable; set to **none** to disable).

Process	Schemes
Radiation	gray (Frierson), RRTMGP correlated- k (LW+SW), ML emulator
Convection	Bechtold, Tiedtke, Kain–Fritsch, Zhang–McFarlane, Emanuel, Kuo, unified mass-flux, EDMF, SBM, DCA
Microphysics	Kessler, Sundqvist, Morrison, Seifert–Beheng, Thompson, super-droplet, ML
Cloud fraction	Sundqvist, Xu–Randall
Boundary layer	Louis, Holtslag–Boville, YSU, EDMF, TKE 1.5, CLUBB-lite, Smagorinsky
Gravity-wave drag	Lindzen, Hines, McFarlane, Rayleigh, prognostic spectral, ML

Table S3: Ocean, land and sea-ice schemes.

Component	Schemes
Ocean vertical mixing	KPP, Richardson, constant; Jayne–St-Laurent tidal
Ocean lateral mixing	harmonic, biharmonic, GM–Redi, Visbeck, Leith, backscatter
Ocean grids	lat–lon, Mercator, cubed-sphere C-D, MPAS Voronoi, tripolar (eORCA)
Land	slab (bucket); multi-layer Richards + Johansen thermal; DALEC carbon; snow
Sea ice	thermodynamic slab; EVP/mEVP dynamic; multi-category ITD; ridging, ponds, brine

(Fig. S6). Cloud-resolving RCEMIP³⁴ runs (Fig. S7) self-aggregate convection; warm-rain microphysics matches a super-droplet Golovin coalescence benchmark^{18,35} (Fig. S8). The Lagrangian super-droplet method (SDM) is supplied here as a non-differentiable reference scheme rather than as a trainable component. Its particle-based formulation is closely connected to what is measured in laboratory and field observations of individual droplets, quantities that are otherwise very difficult to integrate at the global modelling scale because of the scale mismatch, which makes it a valuable oracle for calibrating the bulk schemes used in coarser configurations.

S5 Ocean validation

Beyond the main-text comparison (Fig. 5), Fig. S9 shows sea-surface salinity, mixed-layer depth, a temperature–depth snapshot and a salinity section on the MPAS/Voronoi grid against NEMO³⁶, and the MPAS sea-surface-height field. Figure S10 shows a multi-decadal Drake Passage spin-up: the zonal transport and zonal-mean zonal-velocity structure equilibrate, demonstrating a stable Antarctic Circumpolar Current under the split-explicit solver.

S6 Performance and scaling

The single-GPU acceleration numbers below were measured on one consumer-class laptop GPU (NVIDIA RTX, thermally limited) plus a 24-core CPU; the resolution-resolved scaling of Figs. S11 and S12 was measured on NVIDIA A100 GPUs and on a CPU cluster with one MPI process per core. Headline single-GPU throughput (Mcells s^{-1} , fp32) is 678 (lat–lon atmosphere), 404 (cubed-sphere), 298 (icosahedral), 400 (lat–lon ocean), 313 (tripolar) and 252 (MPAS ocean); GPU acceleration over the CPU is 16–21 $\times$ (atmosphere) and 33–99 $\times$ (ocean) at fp32. Simulated-years-per-day (SYPD) and cell throughput as functions of MPI-rank count (CPU) and A100 count are

shown per grid and resolution in Figs. S11 and S12: a single A100 outperforms a full CPU node by roughly one to two orders of magnitude, and throughput keeps rising with device count until the per-device tile becomes communication-bound. A halo-batching change (fewer, larger messages) improves the MPAS halo path (Fig. S13).

Reverse-mode differentiability is preserved across distributed execution: gradients through MPI halo exchange (`sendrecv` with a custom adjoint) and `allreduce`(SUM) match single-rank references to $\sim 10^{-9}$. Documented limits: spectral transforms are communication-bound on multi-GPU, and cross-node cubed-sphere panel exchange anti-scales on a commodity TCP fabric (these are capability rather than speed-up results).

S7 Accessibility: a worked example

Because the model is idiomatic Python, configuring an experiment (here a single column with gray radiation and a Louis boundary-layer scheme) is a few readable lines, and the same physics object can be wrapped by a neural network or differentiated. This low barrier is a core design goal for teaching and rapid prototyping.

```
from legoesm.atmosphere.scm import SingleColumnModel
from legoesm.atmosphere.physics import PhysicsConfig, RadiationConfig, TurbulenceConfig
import jax.numpy as jnp

scm = SingleColumnModel.create(
    physics_config=PhysicsConfig(
        radiation=RadiationConfig(scheme="gray"),
        turbulence=TurbulenceConfig(scheme="louis")),
    nlev=40, dt=300.0,
    T_profile=jnp.linspace(220.0, 295.0, 40), latitude_deg=0.0)

final_state, history = scm.run(nsteps=288, save_every=12)  # 24 h at dt=300 s
```

Swapping `scheme="gray"` for `"rrtmgp"`, or `turbulence` for an `"edmf"` or ML scheme, requires changing only the configuration; differentiating the run with respect to any parameter requires wrapping `scm.run` in `jax.grad`.

S8 Physical consistency: operator-level tests

Operators are verified against closed-form solutions (Fig. S14). The vector-invariant advection reproduces the Burgers tendency $-u \partial_x u$; the Laplacian viscous term reproduces $A_h \partial_{xx} u$ and its k^2 spectral scaling and exponential decay rate; the Coriolis term gives $dv/dt = -fU$, $du/dt = +fV$; and free inertial oscillations trace the analytic hodograph. Under CFL-controlled refinement, finite-volume transport converges at its design order and the semi-implicit gravity-wave solver at second order (Fig. S15); documented gaps (e.g. a parabolic-CFL regime where $\Delta t \propto \Delta x$) are reported rather than masked.

RCE SCM scheme sweep: $T(p)$ vs surface moist adiabat

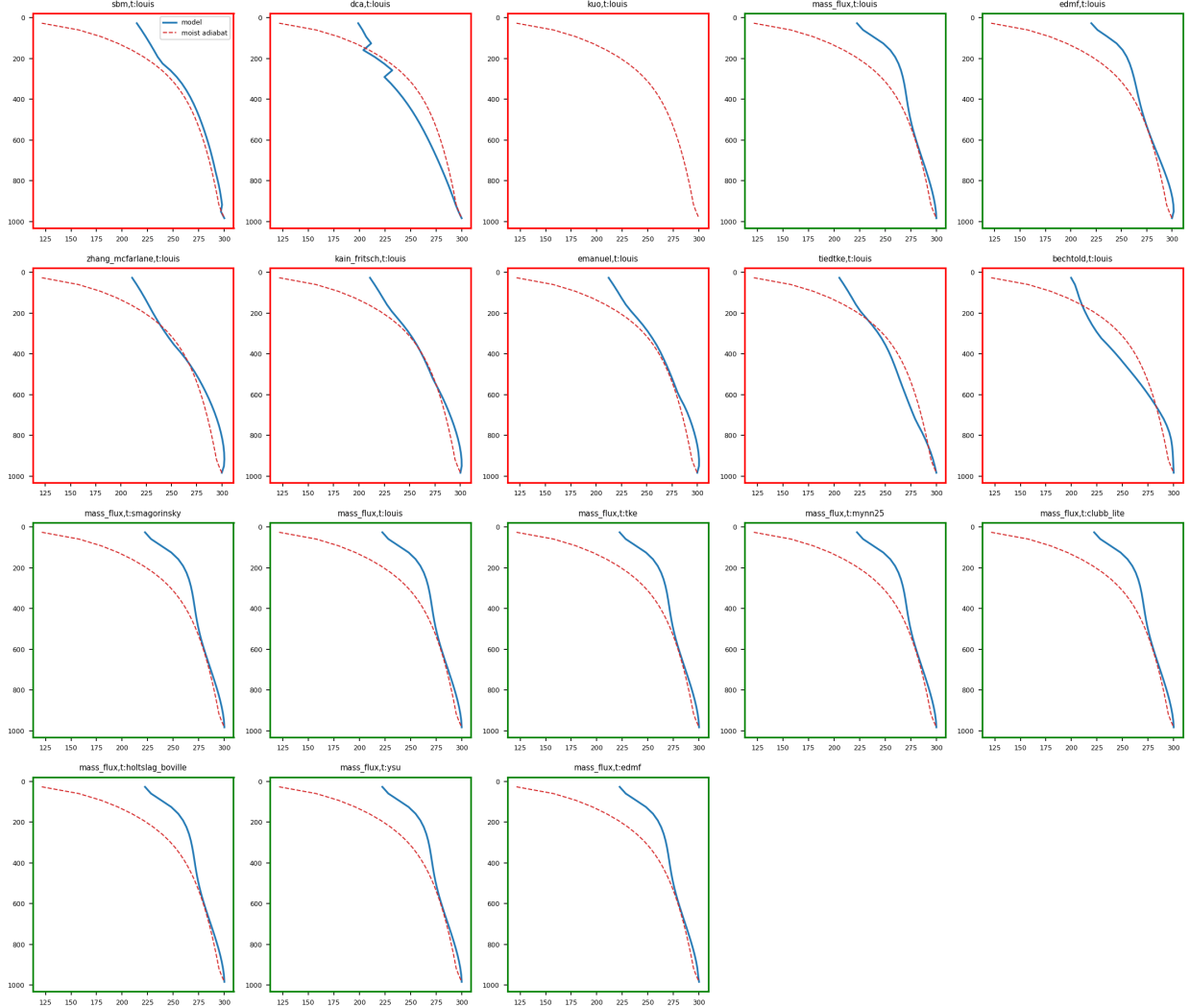


Figure S5: **Full convection-scheme sweep in single-column RCE.** Each panel is one convection parameterization run, unchanged, in a single atmospheric column held in radiative-convective equilibrium: the equilibrated temperature profile (solid) against the surface moist adiabat (dashed), pressure on the vertical axis. Green frames mark schemes that pass the equilibrium-realism gate, red frames those that do not. The lower rows sweep the boundary-layer turbulence closure under a fixed mass-flux convection scheme, showing that the same composability applies across process families. The main-text ranking (main-text Fig. 3) reduces this sweep to a single skill score per scheme.

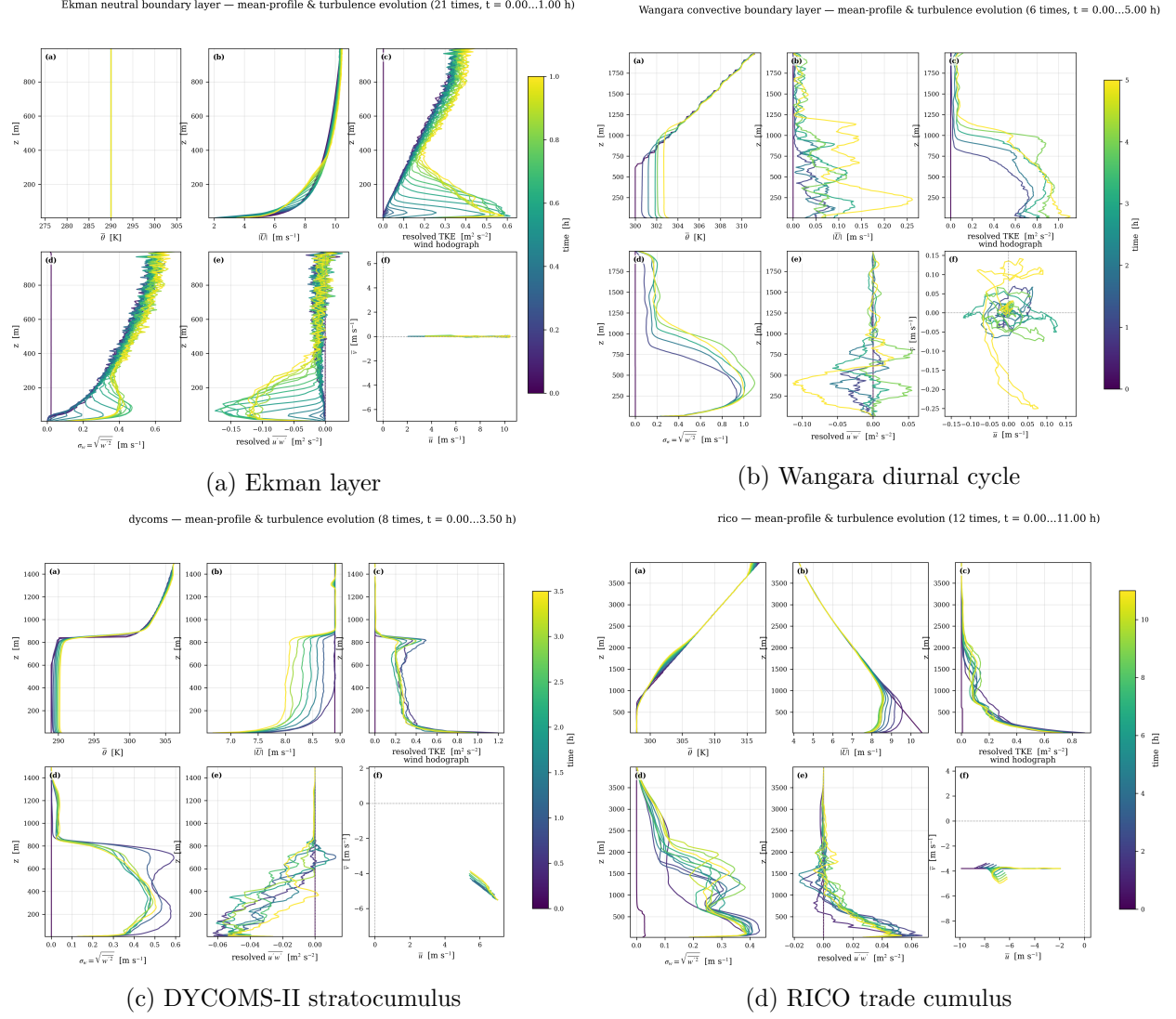


Figure S6: Large-eddy simulation of canonical boundary layers (profile evolution). Together with GABLS1 (main-text Fig. 4a) these span stable, neutral, diurnal, stratocumulus and shallow-cumulus regimes.

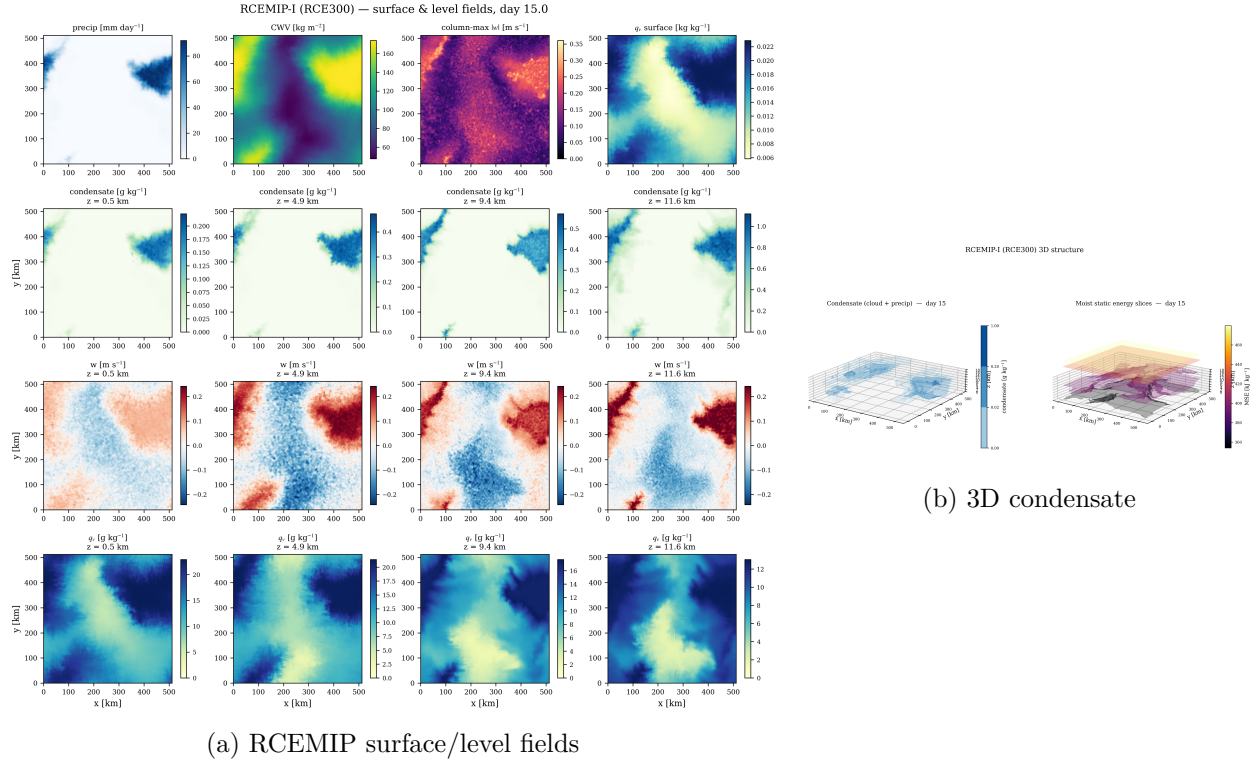


Figure S7: Cloud-resolving RCE (RCEMIP RCE300): self-aggregated convection in surface/level fields and a three-dimensional condensate snapshot.

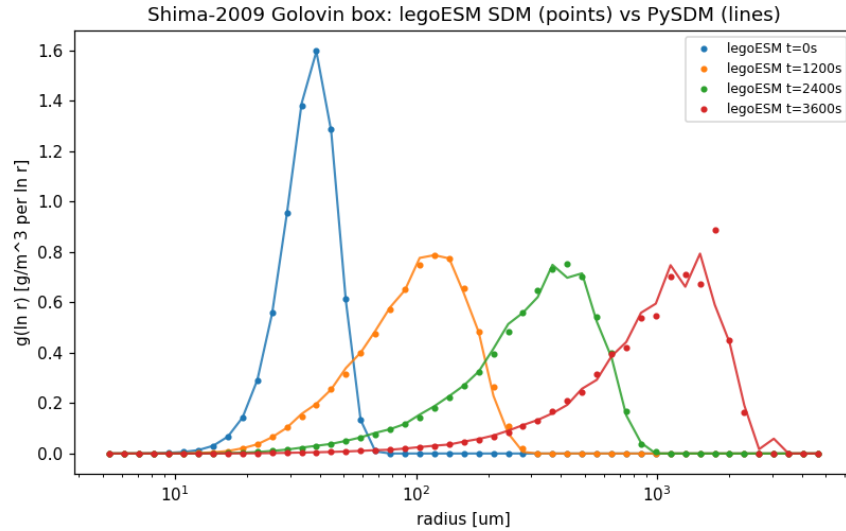
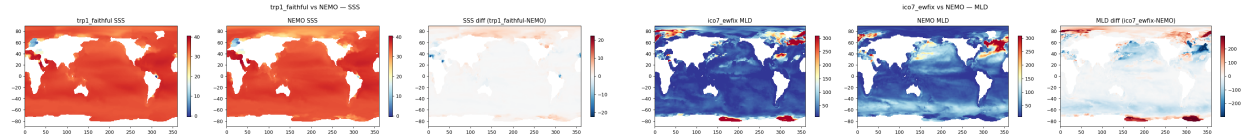
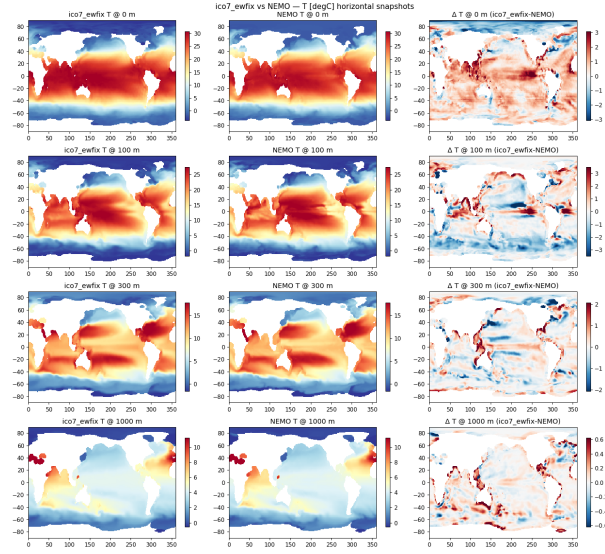


Figure S8: Warm-rain collision-coalescence: legoESM bin/super-droplet microphysics vs. the PySDM super-droplet reference on the Golovin analytic benchmark.

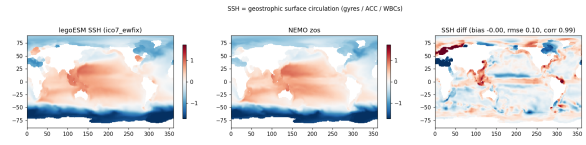


(a) Sea-surface salinity vs. NEMO

(b) Mixed-layer depth vs. NEMO

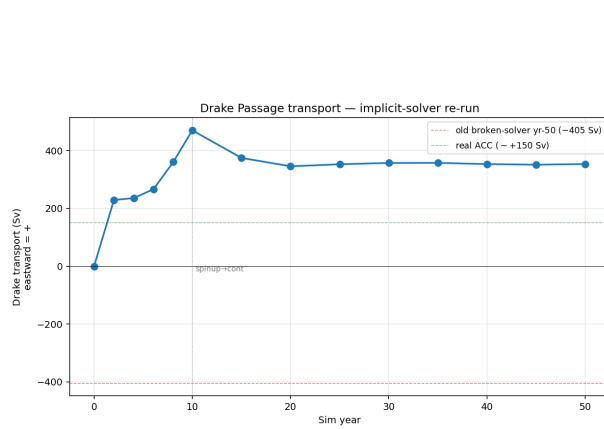


(c) Temperature at depth

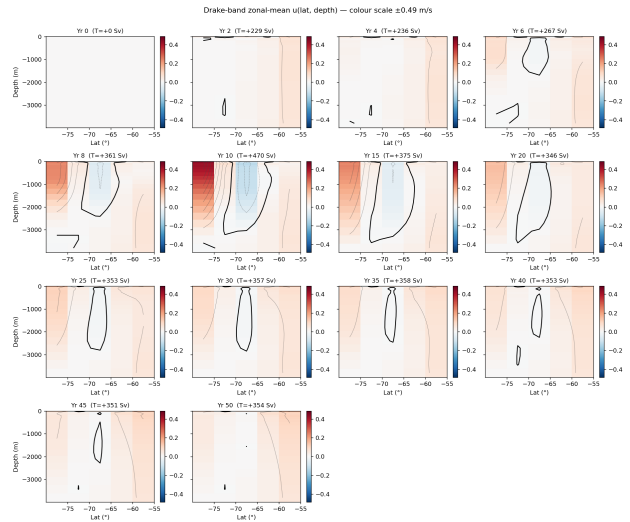


(d) MPAS sea-surface height

Figure S9: Additional MPAS/Voronoi ocean diagnostics against NEMO.



(a) Drake Passage transport



(b) Zonal-mean zonal velocity

Figure S10: Multi-decadal Drake Passage / ACC spin-up showing a stable equilibrated circulation.

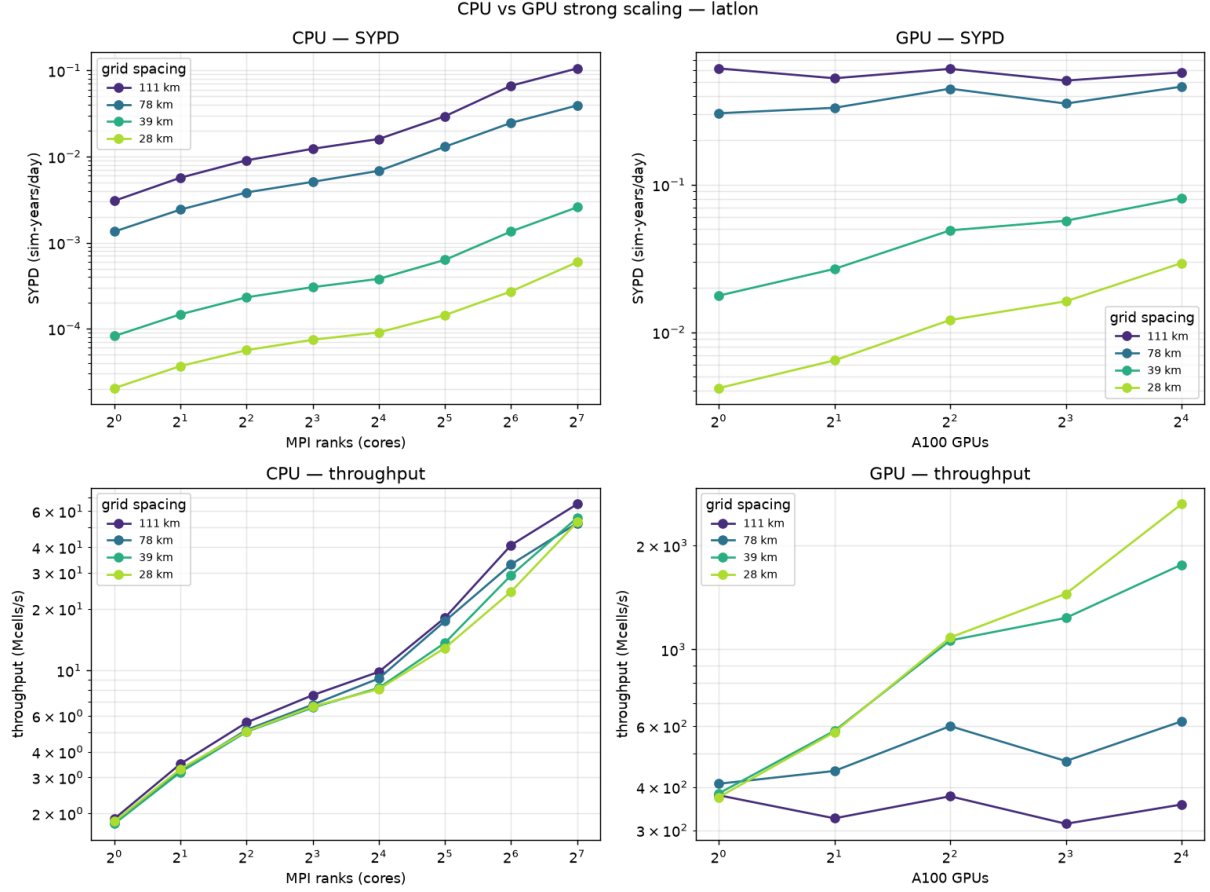


Figure S11: **Lat–lon scaling, CPU vs. A100 GPU**. Simulated-years-per-day (top) and cell throughput (bottom) as a function of MPI ranks (CPU, left) and A100 GPUs (right), at grid spacings from 111 to 28 km. Finer grids need proportionally more devices for the same wall-clock, and a single A100 matches many tens of CPU cores.

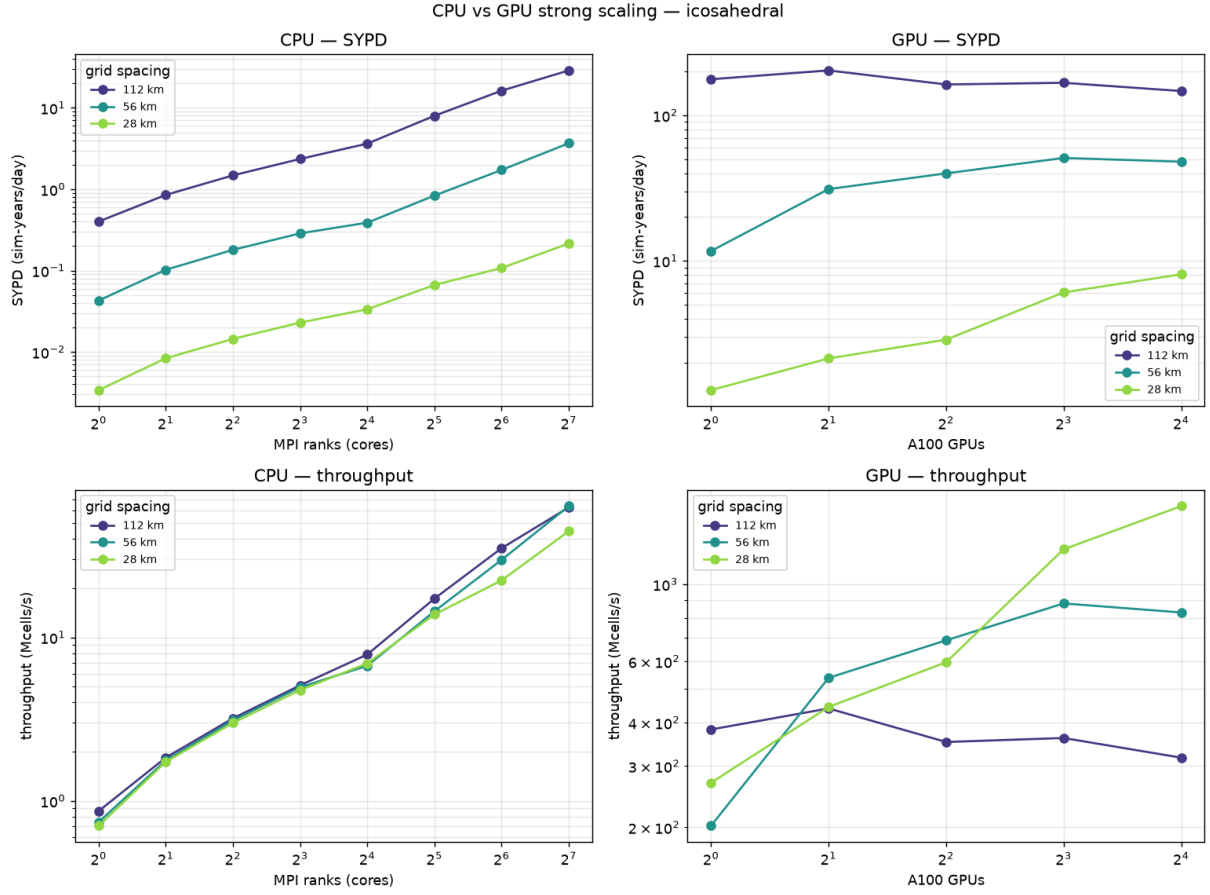


Figure S12: **Icosahedral (MPAS/Voronoi) scaling, CPU vs. A100 GPU.** As Fig. S11, for the icosahedral grid at 112, 56 and 28 km. The unstructured halo yields lower absolute throughput than the structured lat–lon grid at matched resolution, as expected from its memory-access pattern.

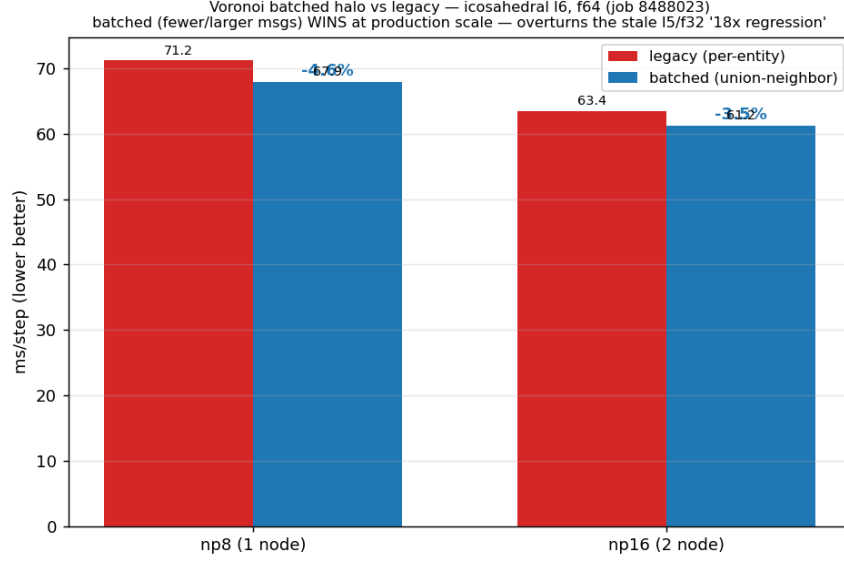


Figure S13: Batched vs. legacy halo exchange on the MPAS/Voronoi grid (icosahedral level-6, fp64): fewer/larger messages reduce time per step at 8 and 16 ranks.

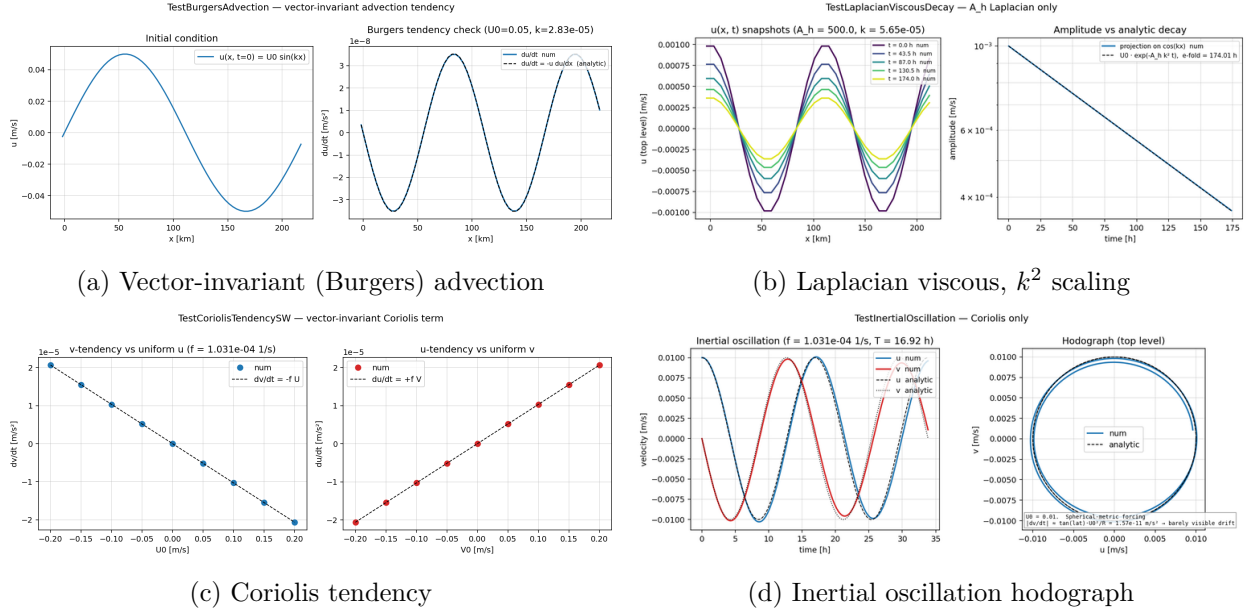
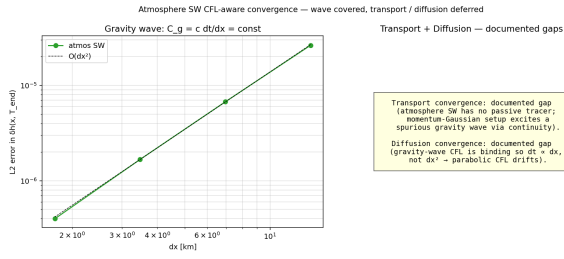
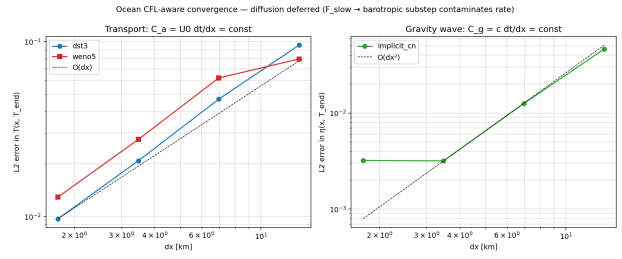


Figure S14: Operator tendencies (solid) against analytic solutions (dashed).



(a) Atmosphere SW



(b) Ocean

Figure S15: CFL-aware convergence: gravity-wave and transport errors vs. grid spacing with design-order reference slopes.

References

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